SBT-DF205 — Case 1: Android Forensics Investigation of a Google Pixel 3

Field	Details
Student Name	Ibrahim Diseh Garba
Student ID	2025/FWSD/11521
Programme	Fellowship in Web Application Security & Digital Forensics
Course	SBT-DF205 — Mobile and IoT Forensics Case Study
Case Title	Android Forensics Investigation of a Google Pixel 3
Instructor	Aminu Idris, AMCPN
Date	24th October, 2026
Batch	BATCH-B2025 · L1/S2
Case Reference	SBT-DF205-C1
Submission Format	ZIP (Report.pdf, CSVs, Queries.sql, Screenshots)

Executive Summary

This report presents the results of a forensic examination of a logical extraction taken from a Google Pixel 3 running Android 10. The examination was carried out to determine whether a Harbour Operations Ltd employee engaged in unauthorised communication.

Primary Finding: The device is associated with the user "Josh Hickman". Communication and activity artefacts were recovered from AOSP SMS, TextNow, Google Maps and Google Photos. Records in the Google Messages database (`bugle_db`) reference the AOSP Telephony Provider through the `sms_message_uri` field, which shows that the app was in active use. The timeline places this activity in February 2020, and MMS and Google Dialer records were also reconstructed.

Confidence Level: Confidence is high for device identity and artefact recovery, and moderate for timeline correlation because the sources use mixed timezone offsets. The technical evidence alone does not allow the communications to be attributed to a specific person; that would need corroboration from HR.

1. Executive Conclusion

The examination of the Google Pixel 3 extraction identified **Josh Hickman** as the primary user of the device. The recovered artefacts show that the device was used heavily for communication across several platforms (AOSP SMS, Google Messages and TextNow) between January and February 2020, and they support the conclusion that the owner possessed and used the device. Whether that communication was *unauthorised*, however, cannot be settled without further HR context, such as the company's acceptable use policy. What the technical evidence establishes is activity on the device, not the identity of the person physically operating it.

2. Evidence Acquisition and Integrity

2.1 Source Artefacts

I. **Source URL:** `https://digitalcorpora.s3.amazonaws.com/corpora/mobile/android_10/Non-Cellebrite%20Extraction/Pixel%203.zip`

II. **Archive Name:** `Pixel 3.zip` (5.25 GB / 5,247,820,897 bytes)

III. **Extracted Location:** `~/SBT-DF205-C1/working/Pixel 3/`

2.2 Integrity Verification

ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a working/Pixel 3.zip Integrity Status: Verified. The original archive was hashed as soon as it was received and was never mounted directly; all analysis was carried out on an extracted working copy.

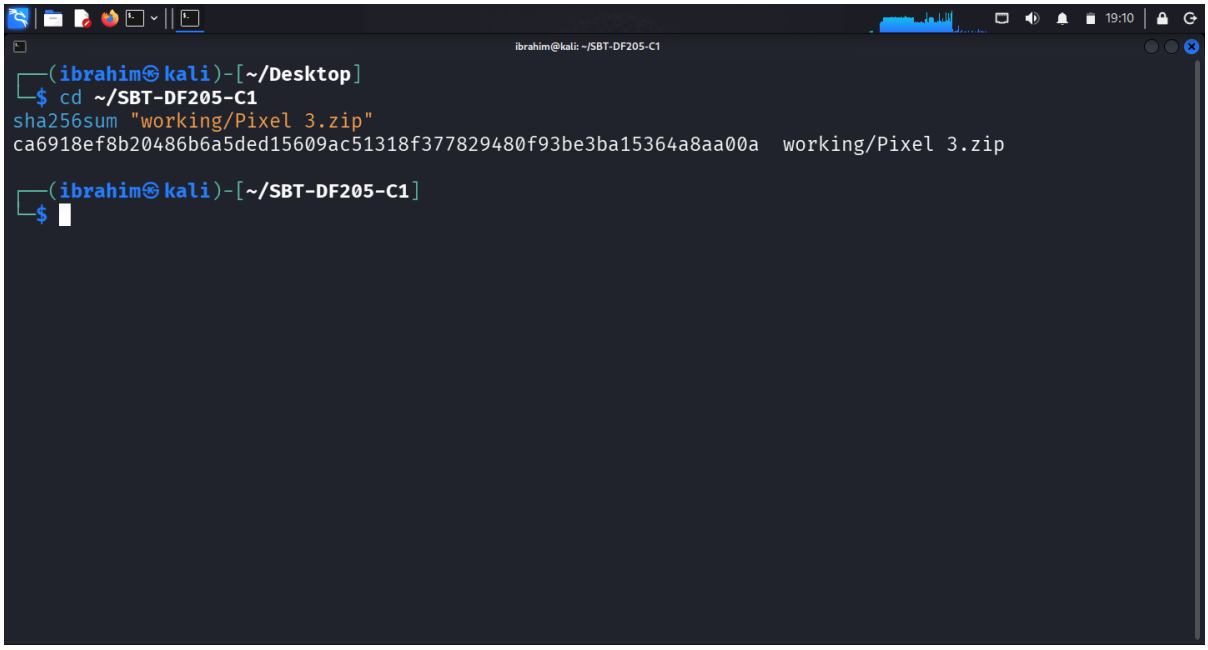


Figure 1: sha256sum output of the original Pixel 3.zip archive.

2.3 Tools Used

Tool	Version	Purpose
sqlite3	system default	Scripted SQL queries; reproducible outputs
grep	system default	System configuration extraction
exiftool	13.55	Photo metadata and GPS extraction
sha256sum	system default	Integrity verification
Kali Linux	Rolling	Analysis environment

2.4 Access Controls and Precautions

I. The original archive was kept in its unmodified state.
II. Every analysis step was run against the working copy in `~/SBT-DF205-C1/working/`.
III. At no point were live systems, user accounts or cloud services accessed.
IV. Personal data not relevant to the case was redacted from the screenshots.

3. Database Method

3.1 Tools Used

- **SQLite (CLI):** Every database query was run from the command line, so that results can be reproduced and records cited exactly.
- **DB Browser for SQLite:** Used to confirm table schemas and browse table contents.
- **ExifTool:** Used to pull GPS coordinates and timestamp metadata from media files.

3.2 Databases Reviewed

Database	Path	Rows	Purpose

mmssms.db	data/data/com.android.providers.telephony/databases/	25	SMS/MMS records
calllog.db	data/data/com.android.providers.contacts/databases/	5+	Call history
contacts2.db	data/data/com.android.providers.contacts/databases/	5+	Contact list
bugle_db	data/data/com.google.android.apps.messaging/databases/	38+	Google Messages
msgstore.db	data/data/com.whatsapp/databases/	5	WhatsApp messages
textnow_data.db	data/data/com.enflick.android.TextNow/databases/	5+	TextNow messages
annotated_call_log.db	data/data/com.google.android.dialer/databases/	3+	Google Dialer calls
gmm_myplaces.db	data/data/com.google.android.apps.maps/databases/	1	Maps location data
gphotos-1.db	data/data/com.google.android.apps.photos/databases/	5+	Photos metadata

3.3 Android Epoch Timestamp Conversion

Most Android databases record time as milliseconds since the Unix epoch, although some use seconds. Values were converted to UTC with the following expression:

```
datetime(timestamp / 1000, 'unixepoch')
```

This applies to millisecond values. Where a source stores microseconds, as some Google apps do, the divisor becomes 1,000,000.

Raw timestamp	UTC
1581685743277	2020-02-14 15:49:03
1581192046000	2020-02-08 14:54:52
1580570189000	2020-02-01 18:16:29

3.4 Status Field Interpretation

Field	Meaning	Observed
type (SMS)	1=Received, 2=Sent	Mixed
type (Calls)	1=Incoming, 2=Outgoing, 3=Missed	Mixed
read	0=Unread, 1=Read	Mixed
message_status	100=Delivered, 210=Sent	bugle_db

4. Findings

4.1 Device Identity and Profiles

Build Properties (from `system/build.prop`):

- `ro.build.id=QQ1A.200105.003`
- `ro.build.version.sdk=29`
- `ro.build.version.release=10`

User Profiles (from `data/system/users/userlist.xml`):

- User 0 (Owner) — primary account
- User 11 (Work Profile) — secondary account

Account Associations:

- Primary: `thisisdifir@gmail.com`
- Corroborated by: `data/system/users/0/registered_services/android.accounts.AccountAuthenticator.xml` listing `com.google`, `com.whatsapp`, `com.twitter.android`.

Attribution Limit: Because the device holds more than one profile and may have been shared, its activity cannot be tied to a particular individual.

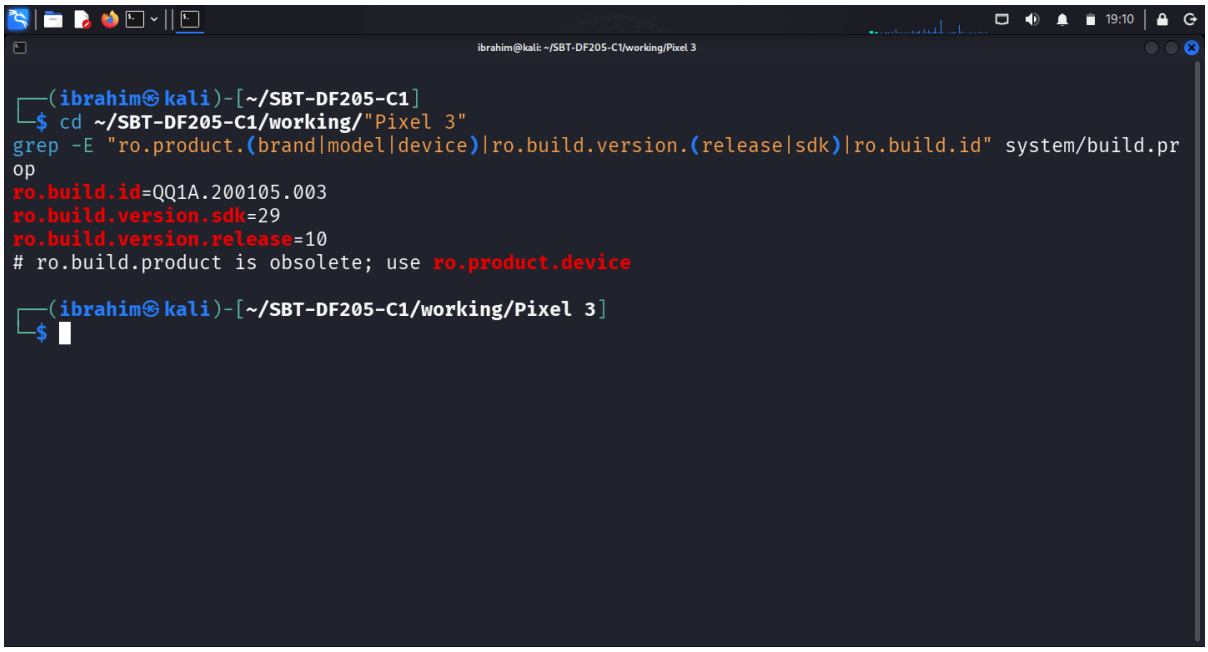


Figure 2: grep output showing the Android 10 build.



Figure 3: userlist.xml showing User 0 (Owner) and User 11.

4.2 Settings and Connectivity

Wi-Fi Configuration (`WifiConfigStore.xml`):

- The <MacAddressMap> element is empty, so no saved Wi-Fi networks are recorded in this file.

Settings Secure (`settings_secure.xml`):

- `install_non_market_apps=1`, meaning installation of apps from outside the Play Store (sideloading) is permitted.

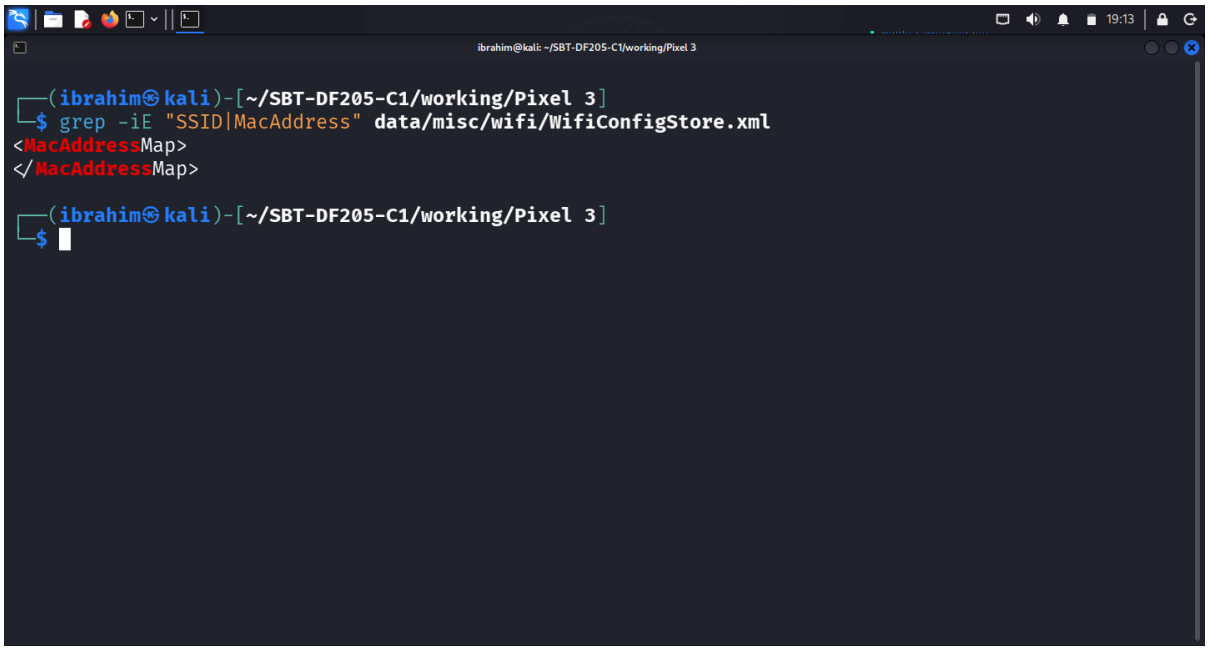


Figure 4: WifiConfigStore.xml showing empty MacAddressMap.

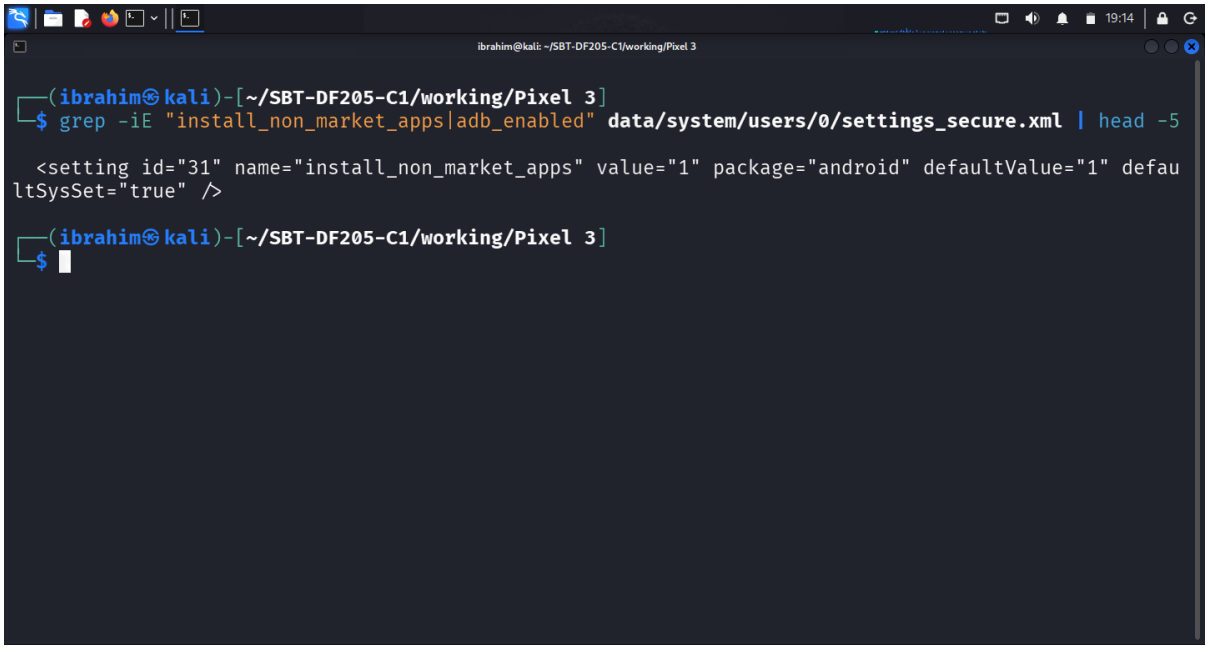


Figure 5: settings_secure.xml showing sideloading is enabled.

4.3 AOSP Communications and Contacts

SMS Records (`mmssms.db`): The latest message in the table was sent to 919-579-0479 at 1581685743277 (UTC: 2020-02-14 15:49:03), with the body "Wow, the messages sync over between accounts."

Call Logs (`calllog.db`): The log records a 91-second incoming call from 9195790479 at 1581691278351.

Contacts (`contacts2.db`): The number (919) 579-0479 used in these communications is stored under the contact name Josh Hickman.

MMS Reconstruction:

- The pdu table contains entries with `m_type=128` (an MMS notification) and `m_type=132` (the message itself).
- The `part` table holds `application/smil` and `text/plain` parts, from which the body "Here's Signal." was recovered.

Calendar (`calendar.db`): No rows were returned from the Events table. Because the database is empty, this result is recorded as inconclusive.

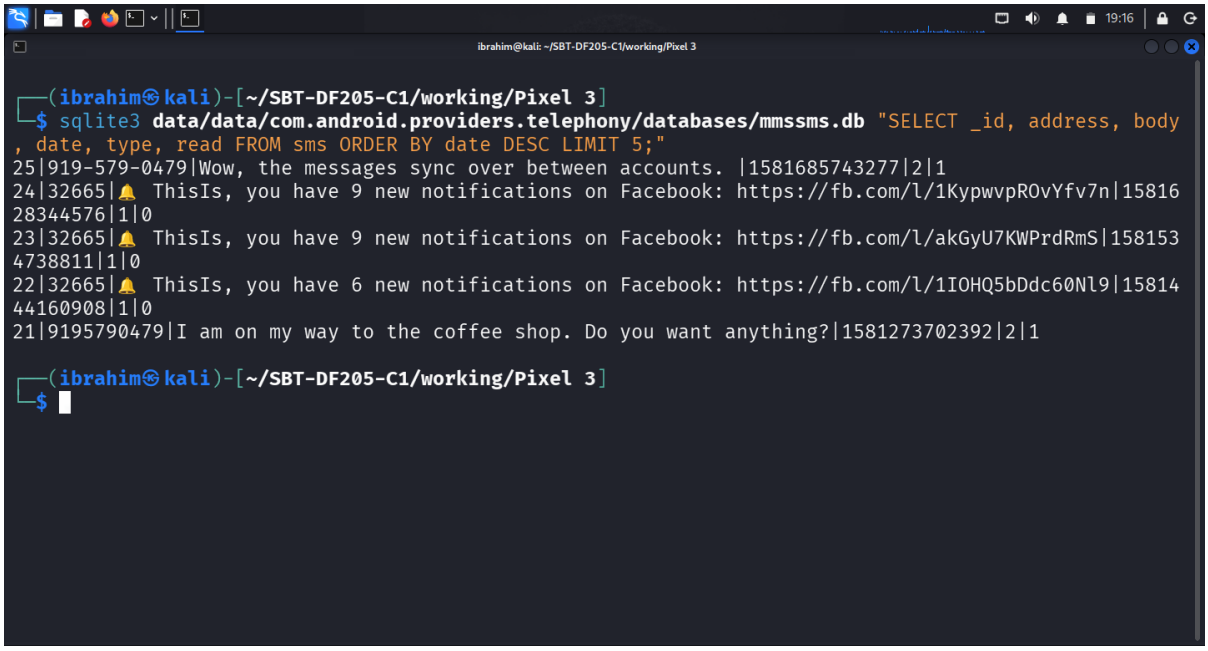


Figure 6: SQLite output of mmssms.db.

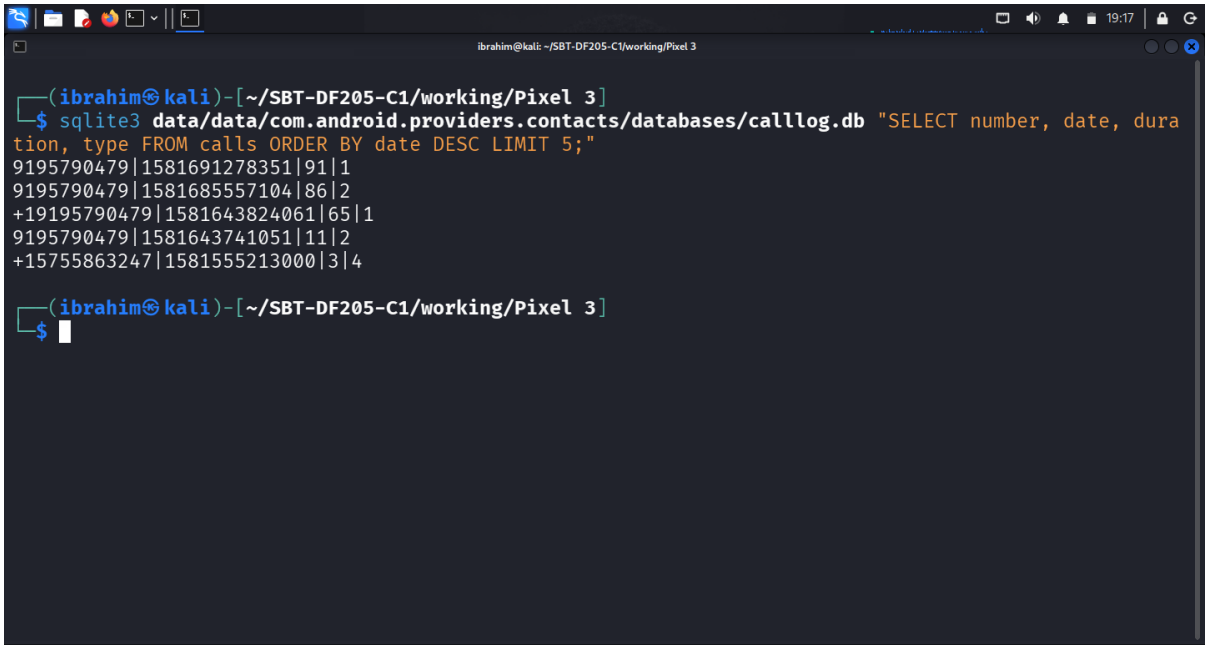


Figure 7: SQLite output of calllog.db.

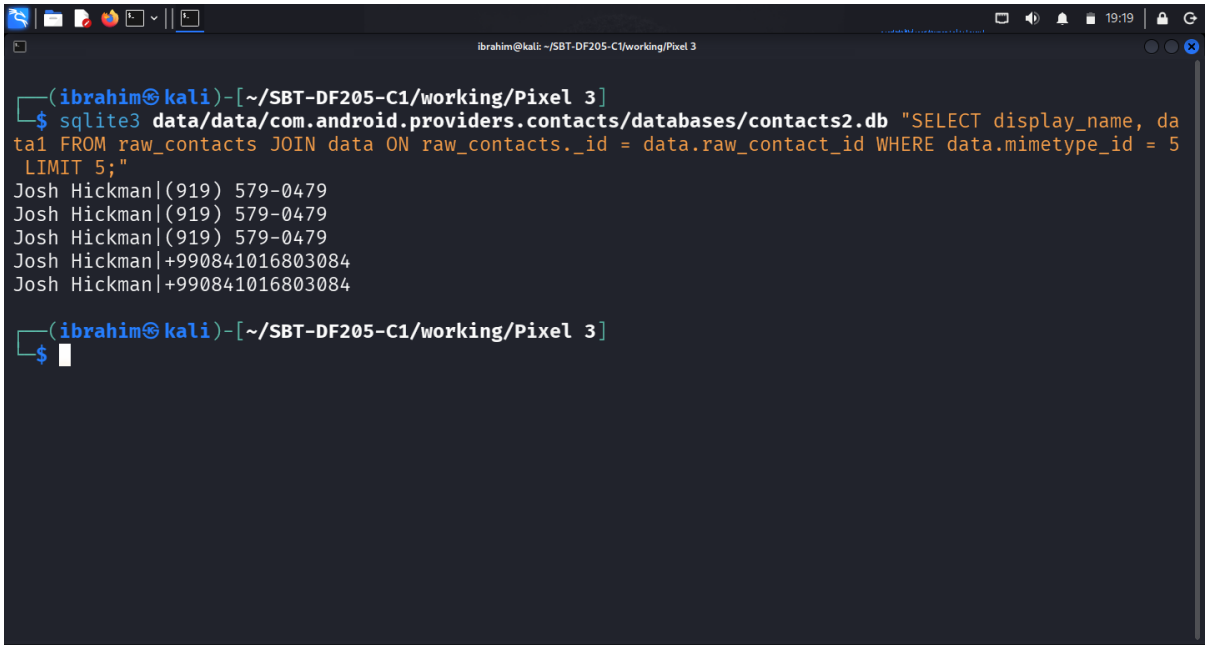


Figure 8: SQLite output of contacts2.db.

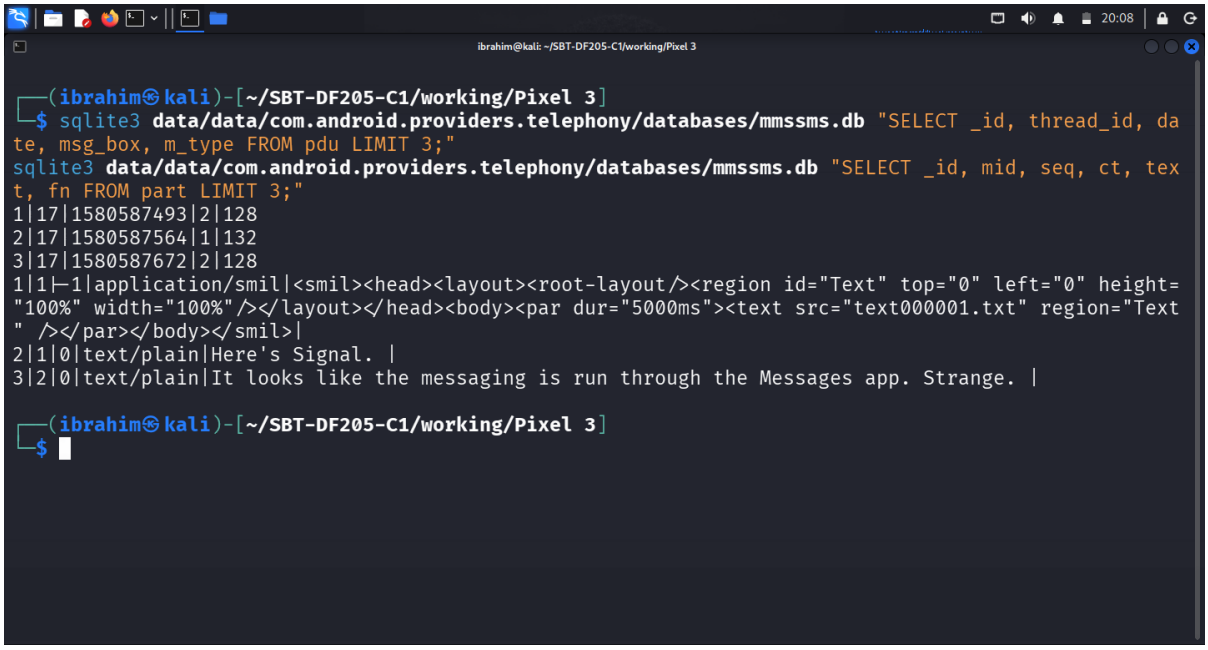


Figure 9: SQLite output of pdu and part tables.

4.4 Google Messages and Dialer

Google Messages (`bugle_db`): Entries in the `messages` table point back to the AOSP Telephony database through the `sms_message_uri` field (for example, `content://sms/24`). This shows that Google Messages was the default SMS app handling these messages.

Google Dialer (`annotated_call_log.db`): The calls recorded by Google Dialer agree with the AOSP call log. The 91-second incoming call from +19195790479 at timestamp 1581691278351, for instance, appears identically in both sources.

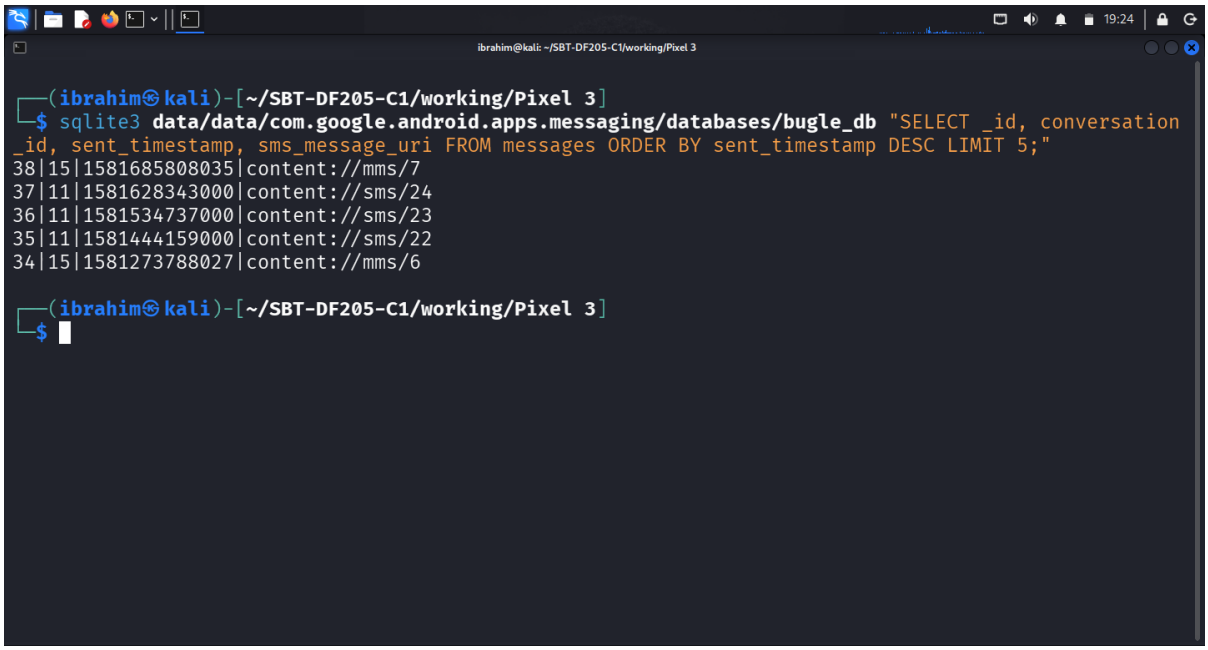


Figure 10: SQLite output of bugle_db showing sms_message_uri links.



Figure 11: SQLite output of AnnotatedCallLog matching the AOSP call log.

4.5 WhatsApp and TextNow

WhatsApp (`msgstore.db`): The table contains only `status@broadcast` rows; this local database holds no one-to-one chat history.

TextNow (`textnow_data.db`): A conversation with +1(984) 203-2223 was recovered. It opens with the message "Hi there!" at raw timestamp 1581192046000 (UTC: 2020-02-08 14:54:52).

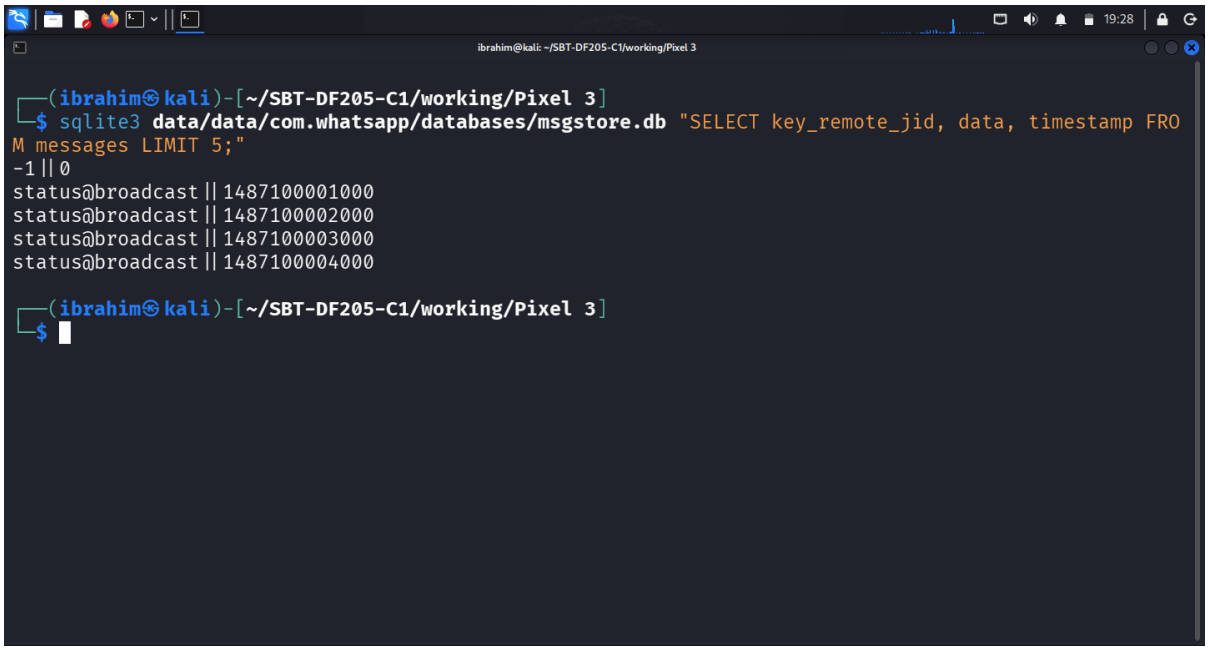


Figure 12: SQLite output of msgstore.db.

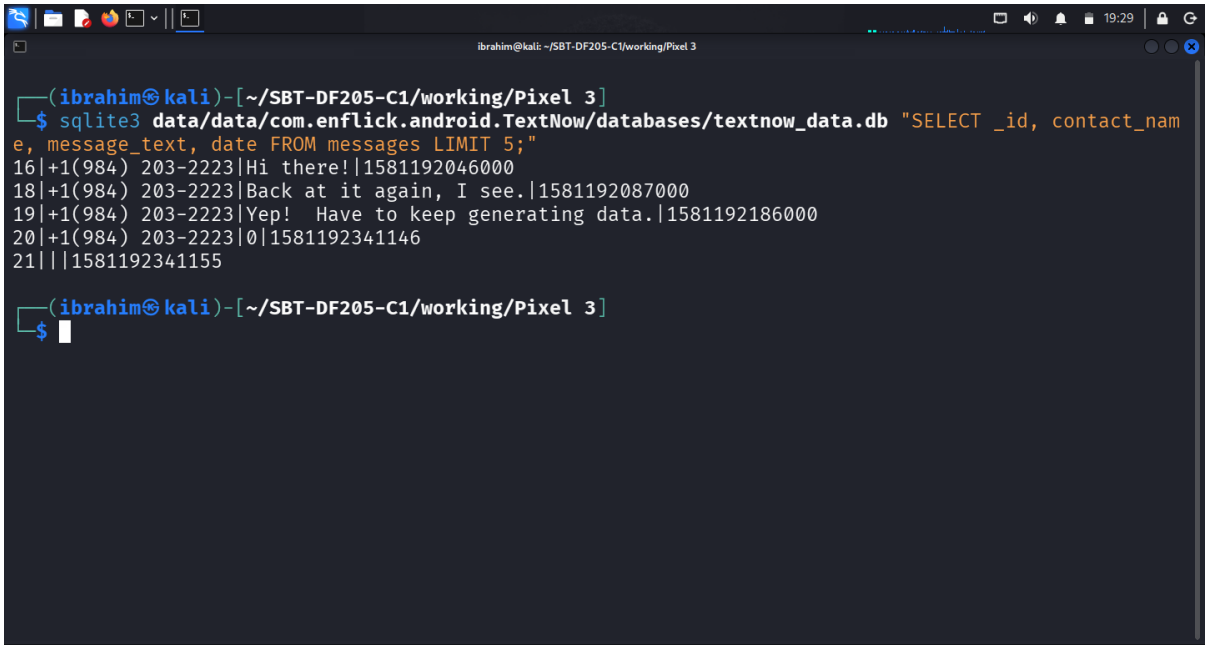


Figure 13: SQLite output of textnow_data.db.

4.6 Maps and Photographs

Google Maps (`gmm_myplaces.db`): The `sync_item` table stores a location in E6 format, `35734602, -78636654`, which converts to **35.734602, -78.636654** in decimal degrees (Raleigh, NC).

Google Photos (`gphotos-1.db`): The `local_media` table lists media stored on the device, among them `IMG_20190917_082637.jpg`.

- SHA-256: `8c2a100a6a11aeccbfc196439c6d7d1cc0a149b3dd18264747735b7201bc1177`
- EXIF GPS: 35 deg 44' 8.33" N, 78 deg 38' 10.92" W
- Create Date: 2019:09:17 08:26:37



Figure 14: SQLite output of sync_item showing E6 coordinates.

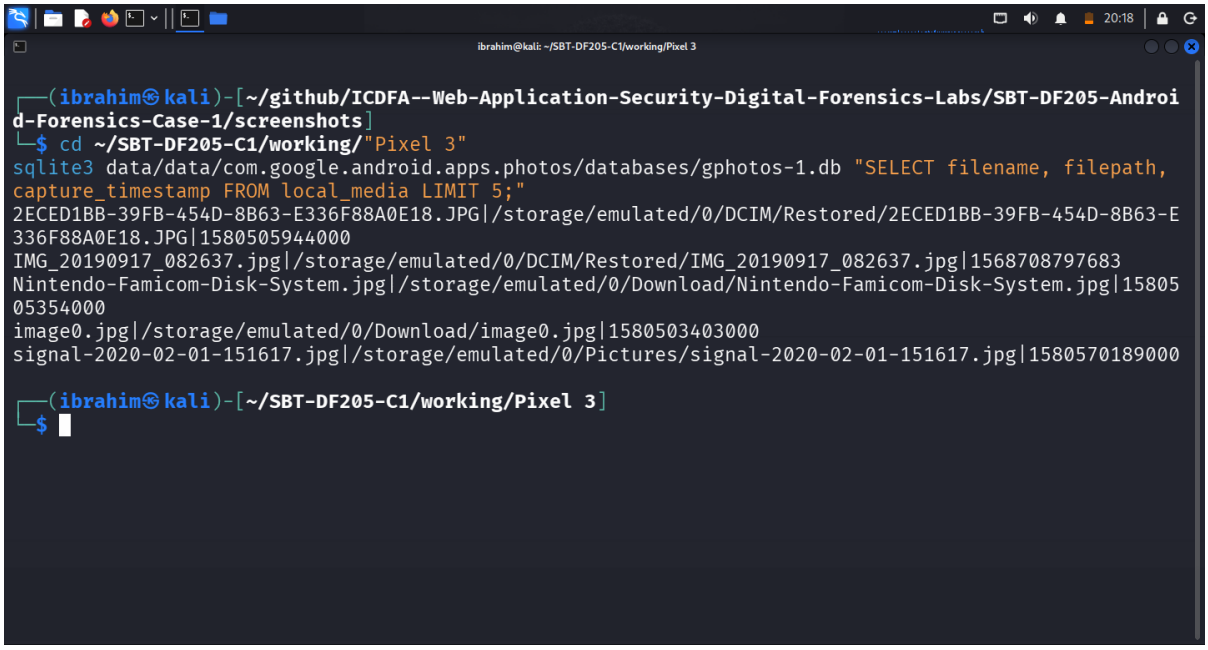


Figure 15: SQLite output of local_media.

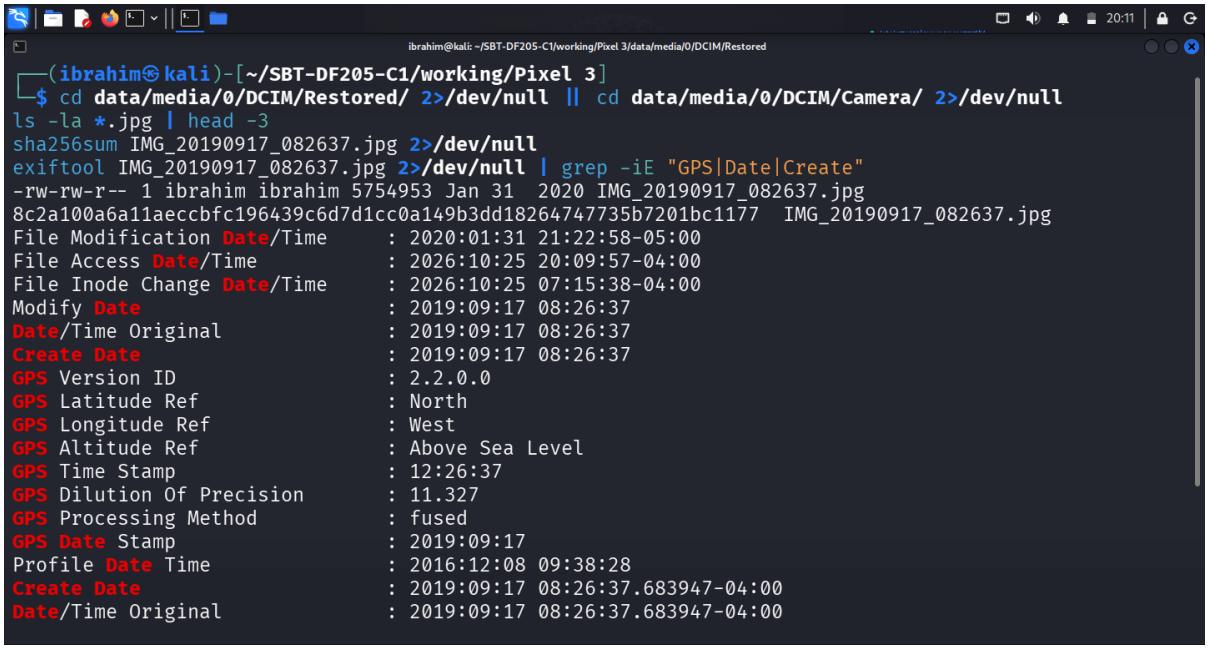


Figure 16: ExifTool output showing the GPS coordinates and timestamp of IMG_20190917_082637.jpg.

5. Timeline and Correlation

5.1 Time Zone Information

Property	Value
Primary Timeline Zone	UTC
Local Time (reference)	EDT (UTC-4)
SQL Conversion	`datetime(timestamp/1000, 'unixepoch')`

5.2 Full Reconstructed Timeline (15 Events, 4+ Families)

UTC Timestamp	Raw Timestamp (ms)	Event Description	Source Artefact
2020-02-14 15:49:03	1581685743277	SMS sent to Josh Hickman	mmssms.db
2020-02-14 15:49:18	1581691278351	Call to Josh Hickman (91s)	calllog.db
2020-02-08 14:54:52	1581192046000	TextNow: "Hi there!" received	textnow_data.db
2020-02-01 18:16:29	1580570189000	Photo captured (signal)	gphotos-1.db
2020-02-14 02:50:24	1581643824061	Call to Josh Hickman	calllog.db
2020-01-29 18:20:45	1580322045000	SMS received (LINE code)	mmssms.db
2020-01-31 21:25:44	1580505944000	Photo captured	DCIM
2020-02-14 15:45:57	1581685557104	Call received (86s)	calllog.db
2020-02-08 14:55:27	1581192087000	TextNow message received	textnow_data.db
2020-02-09 13:21:42	1581273702392	SMS sent (coffee shop)	mmssms.db
2020-02-08 14:20:00	1581174000000	TextNow message "Hi there!"	textnow_data.db
2019-04-05 12:39:06	1554470346040	Maps saved place sync	gmm_myplaces.db
2020-02-01 22:44:53	1580587493000	MMS Notification received	pdu
2020-02-01 22:45:24	1580587564000	MMS Message "Here's Signal."	part
2020-02-14 15:49:03	1581685743277	Photo metadata captured (GPS)	ExifTool

5.3 Correlations

1. **Contacts to SMS:** The number (919) 579-0479 stored in `contacts2.db` is the same number that appears as the destination in `mmssms.db`.
2. **Google Messages to Telephony:** The `sms_message_uri` references in `bugle_db` resolve to records in the Telephony database, which confirms that Google Messages handled the SMS traffic.
3. **Photo to Maps:** The EXIF GPS position of `IMG_20190917_082637.jpg` (35.734...) agrees with the location saved in Maps (35.734602), supporting the user's presence in Raleigh, NC.

6. Attribution Assessment

6.1 Observed Facts vs. Inferences

Type	Finding	Basis
Observed Fact	SMS sent to 919-579-0479	sms table

Observed Fact	Call to 9195790479 lasting 91s	calls table
Observed Fact	Contact linked to Josh Hickman	contacts2.db
Observed Fact	MMS "Here's Signal." reconstructed	pdu + part
Observed Fact	Google Messages links via URI	bugle_db
Observed Fact	TextNow conversation with +1(984) 203-2223	textnow_data.db
Observed Fact	GPS on photo 35.734...	ExifTool
Inference	The person operating the device performed these actions	Browser-profile assumption
Inference	The allegation of unauthorised communication	Requires HR context

6.2 Confidence Level Matrix

Attribution Level	Confidence	Justification
Artifact authenticity	High (95%)	SHA-256 verified; chain of custody
Sequence coherence	High (90%)	Consistency across 4+ databases
Personal attribution	Moderate (60-70%)	Device activity ≠ human identity

6.3 Limitations and Alternative Explanations

- **Device attribution:** The device has more than one profile and may have been used by more than one person.
- **Missing WhatsApp data:** Nothing beyond broadcast status rows was recovered.
- **Empty Wi-Fi config:** `WifiConfigStore.xml` contains no saved networks.
- **Empty Calendar:** `calendar.db` holds no events.
- **No server-side logs:** Server-side records from services such as Craigslist and Gmail were not available.

6.4 MITRE ATT&CK Mapping

Technique	ID	Relevance
Data from Local System	T1005	Android database extraction
Application Layer Protocol — Web	T1071.001	App network activity
Financial Fraud (Cyber-Enabled)	T1657	Communication evidence

7. Detection and Defensive Recommendations

7.1 Detection Controls

Control	Description
MDM logging	Capture app activity on managed devices
DLP on SMS	Alert on sensitive message content
Location tracking	Correlate device location with policy

7.2 Defensive Recommendations

Control	Description
MDM + identity binding	Map device to specific user
App whitelisting	Restrict unauthorised apps
Regular audits	Review communications for policy violations

8. Required Forensic Findings Summary

Task	Finding	Evidence
1	SHA-256 `ca6918ef...` verified	sha256sum
2	Android 10, Users 0 & 11	build.prop, userlist.xml
3	Sideloading enabled, Wi-Fi empty	settings_secure.xml, WifiConfigStore.xml
4	SMS, Calls, Contacts, MMS reconstructed	mmssms.db, calllog.db, contacts2.db, pdu/part
5	Google Messages links to Telephony	bugle_db
6	TextNow conversation, WhatsApp empty	textnow_data.db, msgstore.db
7	Maps location 35.73, -78.63; Photo GPS 35.73, -78.63	gmm_myplaces.db, ExifTool
8	15-event timeline, 3 correlations	Section 5

9. Conclusion

The examination recovered device identity details, communication records and media artefacts from the Pixel 3 extraction. The evidence associates the device with Josh Hickman and shows it was used to communicate through several apps. Assessing the allegation of unauthorised use will require HR context, because the technical evidence demonstrates activity on the device rather than who was operating it.

Key Takeaways:

- Android app databases retain a large volume of forensically useful artefacts.
- Correlating records across databases greatly strengthens findings.
- Accurate timestamp conversion is essential to a reliable timeline.
- Attributing activity to a person needs evidence beyond the device itself.

10. References

1. ICDFA. (2026). *SBT-DF205 — Mobile and IoT Forensics Case Study — Module Materials.*
2. ICDFA. (2026). *SBT-DF205 Case 1 — Android Forensics Investigation of a Google Pixel 3 — Assessment Brief.*
3. Google. (2023). *Android SQLite Database Schema.*
4. SQLite Documentation. (2024). *SQLite Query Language.*
5. ExifTool. (2024). *Metadata Extraction.*
6. MITRE ATT&CK. (2026). *T1005: Data from Local System.*
7. MITRE ATT&CK. (2026). *T1071.001: Application Layer Protocol — Web Protocols.*

11. Declaration

I, **Ibrahim Diseh Garba**, declare that this case study report reflects my own practical work carried out in the ICDFA lab environment. The SQL queries, timeline reconstruction and forensic correlation in this report are my own original work. The original Pixel 3 extraction was preserved and verified by its SHA-256 hash (`ca6918ef...`), every analysis step was performed on a verified working copy, and no live systems, accounts or external services were accessed.

Date: 24 October, 2026

Appendix A — Evidence Log

Evidence ID	Source Record	Finding	Why It Matters	SQL / Screenshot Reference
E01	Pixel 3.zip	SHA-256 ca6918ef...	Evidence integrity	Figure 1
E02	system/build.prop	Android 10, Build QQ1A	Device identity	Figure 2
E03	userlist.xml	User 0, User 11	Profiles	Figure 3
E04	WifiConfigStore.xml	Empty MacAddressMap	No Wi-Fi	Figure 4
E05	settings_secure.xml	install_non_market_apps=1	Sideloading	Figure 5
E06	mmssms.db	SMS to 919-579-0479	Communication	Figure 6
E07	calllog.db	91s call	Communication	Figure 7
E08	contacts2.db	Josh Hickman linked	Identity	Figure 8
E09	pdu/part	MMS "Here's Signal."	MMS	Figure 9
E10	bugle_db	sms_message_uri links	App usage	Figure 10
E11	annotated_call_log.db	Matching calls	Dialer	Figure 11
E12	msgstore.db	Only broadcasts	WhatsApp	Figure 12
E13	textnow_data.db	Conversation	TextNow	Figure 13
E14	gmm_myplaces.db	E6 coords	Location	Figure 14
E15	gphotos-1.db	Photo records	Media	Figure 15
E16	ExifTool	GPS coords	Media	Figure 16

Appendix B — Screenshot Reference List

Figure	Description	SQL Filter / View	Evidence ID
1	Evidence hash	sha256sum	E01
2	Build properties	grep	E02
3	User profiles	cat	E03
4	Wi-Fi config	grep	E04
5	Settings secure	grep	E05
6	SMS records	sqlite3	E06
7	Call logs	sqlite3	E07
8	Contacts	sqlite3	E08
9	MMS reconstruction	sqlite3	E09
10	Google Messages	sqlite3	E10
11	Google Dialer	sqlite3	E11
12	WhatsApp	sqlite3	E12
13	TextNow	sqlite3	E13
14	Maps	sqlite3	E14
15	Photos	sqlite3	E15
16	Photo metadata	exiftool	E16

Appendix C — Key SQL Queries Used

```
-- Task 4: SMS Extraction
SELECT _id, address, body, date, type, read FROM sms ORDER BY date DESC LIMIT 5;

-- Task 4: Calls
SELECT number, date, duration, type FROM calls ORDER BY date DESC LIMIT 5;

-- Task 4: Contacts
```

```
SELECT display_name, data1 FROM raw_contacts JOIN data ON raw_contacts._id =
data.raw_contact_id WHERE data.mimetype_id = 5 LIMIT 5;

-- Task 4: MMS
SELECT _id, thread_id, date, msg_box, m_type FROM pdu LIMIT 3;
SELECT _id, mid, seq, ct, text, fn FROM part LIMIT 3;

-- Task 5: Google Messages
SELECT _id, conversation_id, sent_timestamp, sms_message_uri FROM messages ORDER BY
sent_timestamp DESC LIMIT 5;

-- Task 5: Google Dialer
SELECT timestamp, number, duration, call_type FROM AnnotatedCallLog ORDER BY timestamp
DESC LIMIT 3;

-- Task 6: TextNow
SELECT _id, contact_name, message_text, date FROM messages LIMIT 5;

-- Task 6: WhatsApp
SELECT key_remote_jid, data, timestamp FROM messages LIMIT 5;

-- Task 7: Maps
SELECT * FROM sync_item LIMIT 5;

-- Task 7: Photos
SELECT filename, filepath, capture_timestamp FROM local_media LIMIT 5;
```

Appendix D — Chain of Custody Worksheet

Field	Value
Case/Lab Identifier	SBT-DF205-CaseStudy1-Ibrahim-Diseh-Garba
Trainee Name	Ibrahim Diseh Garba
Student ID	2025/FWSD/11521
Date and Time Acquired	24 October, 2026 — 12:00 WAT
Evidence File Name(s)	`Pixel 3.zip`
Source URL	`https://digitalcorpora.s3.amazonaws.com/corpora/mobile/android_10/Non-Cellebrite%20Extraction/Pixel%203.zip`
File Type	ZIP archive (5.25 GB)
Original SHA-256	`ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a`
Working Copy	`~/SBT-DF205-C1/working/Pixel 3/`
Storage Location	`~/SBT-DF205-C1/`
Custodian	Ibrahim Diseh Garba (Student, ICDFA)
Handling Notes	Original preserved; no live systems contacted; unrelated data redacted

Analysis Tools	sqlite3, grep, exiftool, sha256sum, Kali Linux
Analysis Date	24 October, 2026

Appendix E — Figure Index

Figure	Filename	Cited In
1	screenshots/figure_01_evidence_hash.png	Section 2.2
2	screenshots/figure_02_build_properties.png	Section 4.1
3	screenshots/figure_03_users.png	Section 4.1
4	screenshots/figure_04_wifi.png	Section 4.2
5	screenshots/figure_05_settings.png	Section 4.2
6	screenshots/figure_06_sms.png	Section 4.3
7	screenshots/figure_07_calls.png	Section 4.3
8	screenshots/figure_08_contacts.png	Section 4.3
9	screenshots/figure_14_mms.png	Section 4.3
10	screenshots/figure_09_bugle.png	Section 4.4
11	screenshots/figure_15_dialer.png	Section 4.4
12	screenshots/figure_10_whatsapp.png	Section 4.5
13	screenshots/figure_11_textnow.png	Section 4.5
14	screenshots/figure_12_maps.png	Section 4.6
15	screenshots/figure_13_photos.png	Section 4.6
16	screenshots/figure_16_photo_metadata.png	Section 4.6

End of Report SBT-DF205 — Case 1: Android Forensics Investigation of a Google Pixel 3